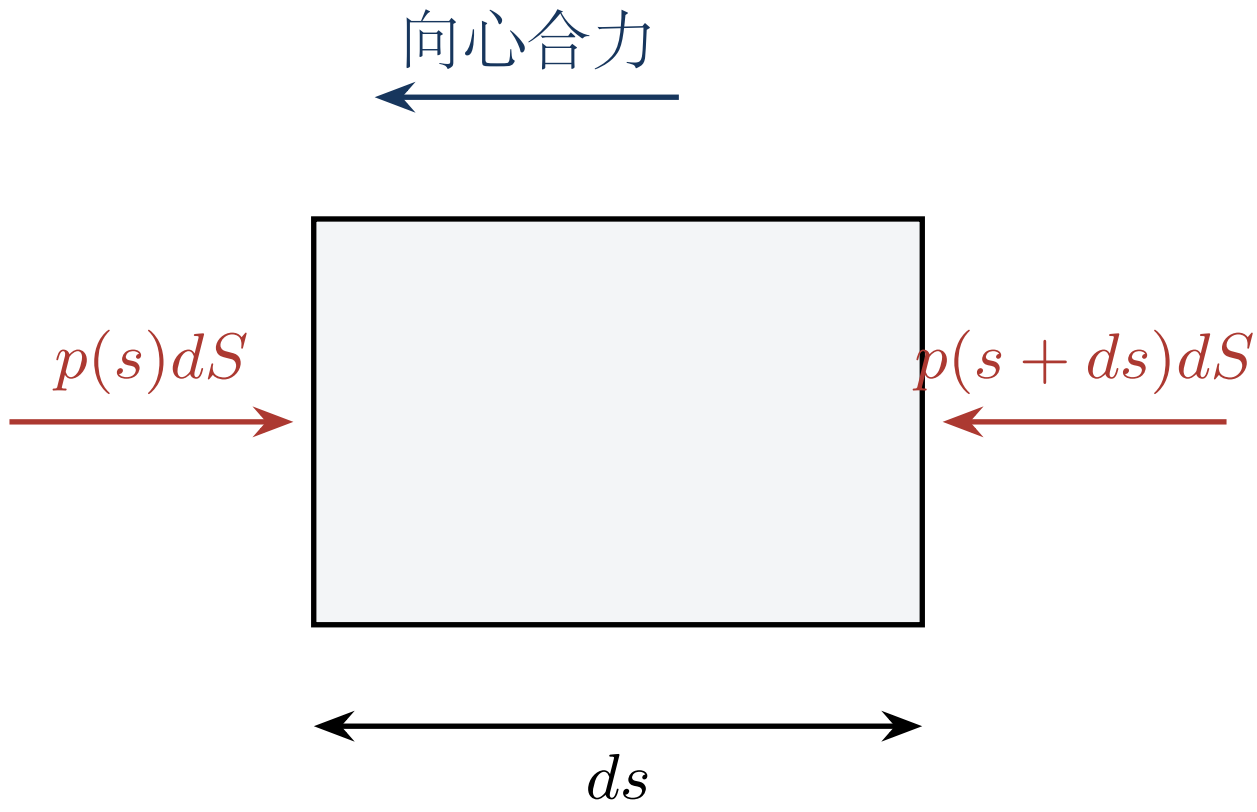

$$[p(s\ +\ ds)\ -\ p(s)]dS\quad =\quad \rho\, dS\, ds\, \omega^2 s$$